Extending Joint Radar-Communications Bounds for FMCW Radar with Doppler Estimation

(Invited Paper)

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***Abstract*—We investigate cooperative in-band radar and communications signaling for frequency-modulated continuous-wave (FMCW) radar and Doppler estimation. While each system typically considers the other system a source of interference, by considering the radar and communications operations to be a single system, joint performance bounds can be formulated. We extend previous work where a novel estimation and information theoretic bound formulation was constructed for a receiver that observes communications and radar returns in the same frequency allocation. While the previous work derived a joint performance bound in terms of the communications rate and the target delay estimation rate of the system for pulsed waveforms, we derive a similar bound for FMCW radar and include Doppler estimation. This extension is important given the rise of popularity of FMCW radars, and since the continuous signaling brings them closer to how communications systems operate.**

I. Introduction

In previous works [1], joint radar estimation and communications Shannon bounds were found in an attempt to define the attainable region for cooperative pulsed radar and communications operation. Here we extend this work by allowing for continuous radar signaling. To facilitate this extension, a Markov motion model and extended measurement model with Doppler estimation are included in the formulation.

A. Background

In [1] and sources therein, the previous approaches to joint radar-communications operations are described as well as the limitations they imposed (mainly isolation in time, frequency, or space).

B. Contributions

In this paper, we extend the results in [1], where a pulsed radar model was used to derive a series of inner bounds on the achievability of the multiple-access channel. The principle contribution of this paper is to derive similar inner bounds for a frequency-modulated continuous-wave (FMCW) radar. The advantages of FMCW radar are numerous, but the range resolution afforded is typically one of the strongest motivators behind its use [2]. While many works in FMCW radar extract the target delay and Doppler via a beat frequency, we assume the more general matched filter. We also introduce another parameter, the coherent processing interval (CPI), which should be tuned to the maximum value to maximize estimation performance, ensuring the period encompasses a stationary target. Due to the complex statistical interdependencies, it is necessary to also include a dynamical Markov model and Doppler estimation into our extended formulation.

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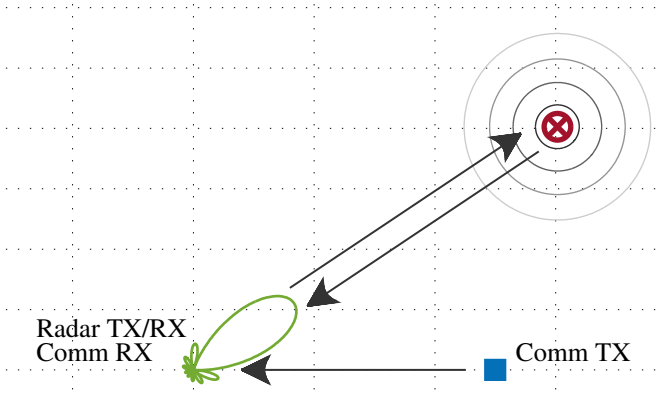


Fig. 1. Main system scenario; target shown in red, communications transmitter node in blue, and radar transceiver pattern in green. The radar acts as the radar transceiver and communications receiver. The uncertainty in the target position is indicated by Gaussian contours.

II. Joint Radar-Communications Model

A radar system functions both as a radar transmitter and receiver to perform target detection and tracking in a monostatic configuration, as well as a communications receiver. The joint system scenario is illustrated in Figure 1. The goal is to derive information theoretic bounds akin to the multiple access bound in communications phenomenology, as discussed in [1] (see Figure 1 within). The information theoretic rates for this model are well known for two communications users [3, 4]. However, it serves only as an introduction and analogy, since one of our "users" is not a communications system attempting to communicate from a countable dictionary, but rather a radar system attempting to illuminate and track a target. For the communications user, this is still a valid model, and the resulting rate bound is given by [3]

$$R_{\text{com}} \leq B \log_2 \left[1 + \frac{b^2 \, P_{\text{com}}}{k_{\text{B}} \, T_{\text{temp}} \, B} \right], \tag{1}$$

where B is our available bandwidth in Hertz, b is the combined communications antenna gain (antenna side lobe gain and

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propagation loss), P_{com} is the total communications transmit power in Watts, k_{B} is the Boltzmann constant in Joules/Kelvin and T_{temp} is the absolute temperature in Kelvin. The combined communications gain is given by $b^2 = G_{\text{com}} c^2 / (2\, R_{\text{TX}}\, \omega_c)^2$, for G_{com} the antenna sidelobe gain, c the speed of electromagnetic waves in meters/second, R_{TX} the range to the communications transmitter in meters, and ω_c the center frequency in radians/second. However, for the radar "user," we must formulate an alternative method of information rate to cast it in this multi-user bound.

A. Radar Estimation Rate

In [1], a novel radar estimation rate metric was developed as a way to quantify radar estimation as an information quantity with respect to time. In this sense, the author was able to splice a radar system into what has traditionally been thought of as a multi-communications user model. The result was a set of joint bounds on communications data information rate and radar estimation rate. The quantity was defined as follows:

$$R_{\text{est}} \leq \frac{\text{I}\left(\underline{\theta} + \underline{n}_\theta; \underline{\theta}\right)}{T} = \frac{\text{h}\left(\underline{\theta} + \underline{n}_\theta\right) - \text{h}\left(\underline{n}_\theta\right)}{T}, \tag{2}$$

where $\text{I}(X;Y)$ is the mutual information between random variables X and Y in bits, $\underline{\theta}$ is a vector of radar parameters, $\underline{n}_\theta$ is the vector of estimation noise (independent from the random radar parameters), T is the CPI, and $\text{h}(X)$ is the differential entropy of X in bits. We can see that, much like the communications counterpart, this is a measure of information rate in bits/second. So where is the information coming from, and who is communicating it? One way to envision this is the target, in the form of entropy (uncertainty) with respect to the radar tracker, is an uncooperative communicator. In this sense, we can view the desired information to be communicated as the target source entropy and the undesired entropy as the total channel noise (receiver and estimation noise).

This metric should satisfy some intuitive properties (and we see it does). For example, as radar transmit power increases, so should the estimation rate. Also, as aggregate channel noise decreases, the estimation rate should increase. If the source entropy is low, then the estimation rate decreases. Interestingly, the coupled relationship of the radar and the target then dictate that the radar should reduce the stimulus rate accordingly. Thus, it should be clear that this measure affects other "users" in the same bandwidth allocation. The higher the estimation rate, we expect more use (typically) of the bandwidth which affects an in-band communications system.

B. Dynamical Model / Source Entropy

While previous work [1] focused only on range measurements for pulsed radar, we extend these results for FMCW radar. However, several underlying simplifications in the previous work no longer apply. For example, with large CPIs typical of FMCW radar [2], range-Doppler coupling and Doppler mismatch become significant factors [5]. As a result, Doppler estimation must be included in the formulation. In addition, previous works started with the delay process modeled as a Gaussian. However, we now have to consider the joint delay-Doppler process, which are in general correlated. This is easy to see, since range as a function of time can be used to estimate the range-rate. Therefore, we must start our formulation at the source, and work through the measurement model to capture the interdependencies. The most natural formulation to that end is to use a dynamical Markov model as a part of a tracking process. We therefore assume a constant velocity, two-dimensional (2D) linear motion model with a Gaussian perturbation acceleration model for our target [6]

$$\underline{s}_{k+1} = \begin{bmatrix} x_{k+1} \\ \dot{x}_{k+1} \\ y_{k+1} \\ \dot{y}_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & T & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & T \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_k \\ \dot{x}_k \\ y_k \\ \dot{y}_k \end{bmatrix} + \mathbf{W}_k, \tag{3}$$

where the $\mathbf{W}_k$ is the process noise with covariance defined as

$$\mathbf{Q} = q_k \begin{bmatrix} \frac{T^3}{3} & \frac{T^2}{2} & 0 & 0 \\ \frac{T^2}{2} & T & 0 & 0 \\ 0 & 0 & \frac{T^3}{3} & \frac{T^2}{2} \\ 0 & 0 & \frac{T^2}{2} & T \end{bmatrix}, \tag{4}$$

x_{k+1} is the target position along the x-axis at discrete time step $k+1$, $\dot{x}_{k+1}$ is the target velocity projected on the x-axis, y_{k+1} is the target position along the y-axis, $\dot{y}_{k+1}$ is the target velocity projected on the y-axis, T is the discrete time step duration of the system (duration between time steps k and $k+1$) as well as the CPI, $\underline{s}_k = \begin{bmatrix} x_k & \dot{x}_k & y_k & \dot{y}_k \end{bmatrix}^{\text{T}}$ is the state vector, and q_k is the process model error intensity. While the measurement space reduces our state dimensionality, the Cartesian model allows motion dynamics to advance uncoupled from the coordinate system.

C. Measurement Model / Measurement Entropy

The observed parameters, after Doppler processing [7], are the range r_k and range-rate $\dot{r}_k$. We assume a narrowband environment such that only a frequency shift (Doppler shift) is induced in the returned waveform, not the more general Doppler time scaling. This is possible under the assumption [8]:

$$T\, B \ll \frac{c}{2\dot{r}}. \tag{5}$$

This means that the range and range-rate which offset the peak of the narrowband cross-ambiguity function at the measurement processor relate to the position and velocity state as follows [8, 9]:

$$r = \sqrt{x^2 + y^2}\ ,\ \dot{r} = \frac{x\,\dot{x} + y\,\dot{y}}{\sqrt{x^2 + y^2}}. \tag{6}$$

While this model may be extended for bearing measurements, it is out of the scope of this paper.

For the measurements, we must solve for the transformed entropy of our source Gaussian state. The range, in general, is Rice distributed [10]. Unfortunately, there is no closed form solution for the differential entropy of a Rician [11]. However, we can compute this entropy using numerical integration or Monte Carlo methods [12]. The range-rate is seen to be

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the sum of products of two correlated Gaussians, normalized by a correlated Rician. It is not obvious what the marginal distribution is. To find the joint entropy, we would need to solve for range-rate entropy conditioned upon the range [3].

We then have to consider measurement noise. If we assume additive white Gaussian noise (AWGN), the already complicated distributions are now convolved with the additive Gaussian distribution. The delay and Doppler measurements originate from the same signal source, the matched filter receiver, and therefore their noise is, in general, correlated [8]. So we must consider the delay and Doppler entropy jointly. The difficulties that are mounting warrant an alternate formulation. We can simplify the analysis by linearizing the measurement model about the predicted state, much like the extended Kalman filter [12]. This results in a time-varying, but linear transform on our Gaussian state vector. An affine transform on a multi-variate Gaussian like our source vector results in another multivariate Gaussian [13]:

$$\underline{s}_{k+1} \sim \mathcal{N}\left(\underline{\mu}, \mathbf{\Sigma}\right) \Longrightarrow \mathbf{C}\,\underline{s}_{k+1} \sim \mathcal{N}\left(\mathbf{C}\,\underline{\mu}, \mathbf{C}\,\mathbf{\Sigma}\,\mathbf{C}^{\mathrm{T}}\right), \quad (7)$$

for $\mathbf{C}$ the linearized transform. Our predictive state $\underline{s}_{k+1}$ is defined in Equation (3) with covariance matrix $\mathbf{\Sigma}$. Here, for simplicity, we assume $\mathbf{\Sigma} = \mathbf{Q}$ defined in Equation (4). Note this implicitly assumes we know the prior state of the target. However, we have not lost generality, as one can easily replace $\mathbf{\Sigma}$ with the predicted covariance advanced by the Kalman equations [12]. To construct our linear measurement transform, we consider R and $\dot{R}$ the measurement functions for the range and range-rate respectively, and the partial derivatives are all to be evaluated at the predicted state. For the range, we get the following:

$$\begin{aligned}\frac{\partial R}{\partial x} &= \frac{\partial}{\partial x}\left[\sqrt{x^2+y^2}\right]\Big|_{x,\dot{x},y,\dot{y}=x_{k+1},\dot{x}_{k+1},y_{k+1},\dot{y}_{k+1}} \\ &= \frac{x_{k+1}}{\sqrt{x_{k+1}^2+y_{k+1}^2}} \\ \frac{\partial R}{\partial y} &= \frac{y_{k+1}}{\sqrt{x_{k+1}^2+y_{k+1}^2}}\ ,\ \frac{\partial R}{\partial \dot{x}} = \frac{\partial R}{\partial \dot{y}} = 0\end{aligned} \quad (8)$$

For the range-rate, we get the following:

$$\begin{aligned}\frac{\partial \dot{R}}{\partial x} &= \frac{\dot{x}_{k+1}}{\sqrt{x_{k+1}^2+y_{k+1}^2}} - \frac{x_{k+1}^2\,\dot{x}_{k+1} + y_{k+1}\,\dot{y}_{k+1}\,x_{k+1}}{\left(\sqrt{x_{k+1}^2+y_{k+1}^2}\right)^3} \\ \frac{\partial \dot{R}}{\partial y} &= \frac{\dot{y}_{k+1}}{\sqrt{x_{k+1}^2+y_{k+1}^2}} - \frac{x_{k+1}\,\dot{x}_{k+1}\,y_{k+1} + y_{k+1}^2\,\dot{y}_{k+1}}{\left(\sqrt{x_{k+1}^2+y_{k+1}^2}\right)^3} \\ \frac{\partial \dot{R}}{\partial \dot{x}} &= \frac{\partial R}{\partial x}\ ,\ \frac{\partial \dot{R}}{\partial \dot{y}} = \frac{\partial R}{\partial y}\end{aligned}$$

Then we can construct our measurement transformation as

$$\mathbf{C} = \begin{bmatrix} \frac{2}{c}\frac{\partial R}{\partial x} & 0 & \frac{2}{c}\frac{\partial R}{\partial y} & 0 \\ \frac{2\omega_c}{c}\frac{\partial \dot{R}}{\partial x} & \frac{2\omega_c}{c}\frac{\partial R}{\partial x} & \frac{2\omega_c}{c}\frac{\partial \dot{R}}{\partial y} & \frac{2\omega_c}{c}\frac{\partial R}{\partial y} \end{bmatrix}, \quad (9)$$

where we have made use of the fact that the time delay is given by $\tau = 2r/c$, and the Doppler shift is given by $\omega_{\mathrm{D}} = 2\omega_c\,\dot{r}/c$. Finally, our AWGN from the estimation process $\underline{n}_{k+1}$ is added (thermal noise and waveform-dependent ambiguity [8]):

$$\mathbf{C}\,\underline{s}_{k+1} + \underline{n}_{k+1} \sim \mathcal{N}\left(\mathbf{C}\,\underline{\mu}, \mathbf{C}\,\mathbf{\Sigma}\,\mathbf{C}^{\mathrm{T}} + \mathbf{J}^{-1}\right). \quad (10)$$

$\mathbf{J}^{-1}$ is the scaled inverse Fisher information matrix (FIM). The FIM is given for a single sample, and so we scale this by the integrated signal-to-noise ratio (SNR) to get the post-processing bound [5]. Since the mean of a multivariate Gaussian does not affect the differential entropy [3], we need only to solve for $|\mathbf{C}\,\mathbf{\Sigma}\,\mathbf{C}^{\mathrm{T}} + \mathbf{J}^{-1}|$, or the determinant of our noisy transformed covariance matrix. Then the differential entropy for the complete linearized measurement model is found to be [3]:

$$\mathrm{h}\left(\tau, \omega_{\mathrm{D}}\right) = \frac{1}{2}\log_2\left[\left(2\pi\,e\right)^2\left|\mathbf{C}\,\mathbf{\Sigma}\,\mathbf{C}^{\mathrm{T}} + \mathbf{J}^{-1}\right|\right]. \quad (11)$$

We now must find the estimation noise of our measurements. We only consider a single target scenario to compare with [1] and because of the complexities with computing the performance with multiple targets in FMCW radars [14]. While traditionally in FMCW radar, the beat frequency is extracted to obtain the range estimate by mixing the radar returns with the continuous transmitter reference [15], we assume correlation with a general waveform, in which case we only are affected by the root mean square (RMS) bandwidth B_{rms} and RMS envelope T_{rms} of the signal [8]. We are assuming the Cramér-Rao lower bound (CRLB) is achievable, and sufficiently high SNR such that we can define the FIM to be [8]

$$\mathbf{J} = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}, \quad (12)$$

where

$$\begin{aligned} J_{11} &= \frac{\mathrm{ISNR}}{\pi}\int_{-\infty}^{\infty}\omega^2 |S_{\mathrm{rad}}(j\omega)|^2 d\omega \\ &= 2T s_{\mathrm{rms}}^2\,\mathrm{ISNR}\left(2\pi\,B_{\mathrm{rms}}\right)^2, \\ J_{22} &= 2\,\mathrm{ISNR}\int_{-\infty}^{\infty} t^2 |s_{\mathrm{rad}}(t)|^2 dt \\ &= 2T\,s_{\mathrm{rms}}^2\,\mathrm{ISNR}\ \left(T_{\mathrm{rms}}\right)^2, \text{ and} \\ J_{12} = J_{21} &= 2\Im\left\{\mathrm{ISNR}\int_{-\infty}^{\infty} t\,s_{\mathrm{rad}}(t)\frac{\partial s_{\mathrm{rad}}^*(t)}{\partial t}dt\right\}, \end{aligned} \quad (13)$$

for $s_{\mathrm{rad}}(t)$ our radar waveform, $S_{\mathrm{rad}}(j\omega)$ the Fourier transform of our radar waveform over the CPI, ISNR the integrated SNR and the waveform finite RMS defined as

$$\mathrm{ISNR} = \frac{T\,B\,a^2\,P_{\mathrm{rad}}}{\sigma_{\mathrm{noise}}^2}\ ,\ s_{\mathrm{rms}} = \sqrt{\frac{1}{T}\int_{-T/2}^{T/2}\left|s_{\mathrm{rad}}(t)\right|^2 dt}, \quad (14)$$

respectively. For the integrated SNR, a is the combined radar antenna gain (antenna main lobe gain, radar cross section, and two-way propagation loss), P_{rad} is the total radar transmit power in Watts, and the receiver thermal noise power in Watts

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is defined as $\sigma^2_{\text{noise}} = k_{\text{B}}\, T_{\text{temp}}\, B$. The combined radar gain is given by

$$a^2 = \frac{G^2_{\text{rad}}\, \text{RCS}\, c^2}{16\pi R^4_{\text{targ}}\, \omega_c^2}, \tag{15}$$

for G_{rad} the main lobe antenna gain relative to an isotropic radiator, RCS the radar cross section, and R_{targ} the range to target. Note we have assumed the baseband waveform is constructed on an axis such that the mean frequency is 0 [8]. In addition, we are assuming a continuously transmitted FMCW waveform that is non-repeating over our CPI, well modeled by a random process. This allows arbitrary selection of the CPI, while still attaining a matched filter. Overlapping CPIs are possible, but this would not affect the estimation rate, since it only amounts to staggered processing at the radar receiver. Note we are also assuming sufficient isolation of the transmit and receive waveforms, and that successive CPI waveforms do not impact the matched filter process significantly. By representing our radar signal in polar form, we can only rewrite the third term as:

$$J_{12,21} = 2\,\text{ISNR} \int_{-\infty}^{\infty} t\, \text{E}\left[f(t)\right] |s_{\text{rad}}(t)|^2 dt, \tag{16}$$

where $f(t)$ is the instantaneous frequency function of our FMCW waveform assumed to be a random process. Since the average of the instantaneous frequency should be 0 (unmodulated signal) for our previous simplification (noting we can always choose the center frequency to satisfy this), this term subsequently is on average 0. Then the scaled inverse FIM, or CRLB becomes:

$$\begin{aligned}\mathbf{J}^{-1} &= \begin{bmatrix} \sigma_\tau^2 & \rho\, \sigma_\tau\, \sigma_\omega \\ \rho\, \sigma_\tau\, \sigma_\omega & \sigma_\omega^2 \end{bmatrix} \\ &= \frac{1}{2T\, s^2_{\text{rms}}\, \text{ISNR}} \begin{bmatrix} (2\pi)^{-2} B_{\text{rms}}^{-2} & 0 \\ 0 & T_{\text{rms}}^{-2} \end{bmatrix}.\end{aligned}$$

Finally, we can complete our radar estimation rate

$$R_{\text{est}} \leq \frac{1}{2T} \log_2 \left[\frac{|\mathbf{C}\, \mathbf{\Sigma}\, \mathbf{C}^{\text{T}} + \mathbf{J}^{-1}(B)|}{|\mathbf{J}^{-1}(B)|} \right], \tag{17}$$

where we have parameterized the scaled inverse FIM $\mathbf{J}^{-1}$ by the bandwidth B since we vary this parameter later.

III. Inner Bounds

As in [1], we solve for a series of inner bounds on joint radar-communications operations to test our new outer bounds.

A. Isolated Sub-band

The simplest inner bound we can derive is the isolated sub-band (ISB) inner bound. This is where we allocate sub-bands from our overall bandwidth B to the communications user and radar user separately. This is traditionally how these systems operate, but it serves to build a basis for comparison with other inner bounds. The bounds are easily seen to be

$$R_{\text{com;ISB}} \leq \alpha\, B \log_2 \left[1 + \frac{b^2 P_{\text{com}}}{k_{\text{B}}\, T_{\text{temp}}\, \alpha\, B} \right] \tag{18}$$

and

$$R_{\text{est;ISB}} \leq \frac{1}{2T} \log_2 \left[\frac{|\mathbf{C}\, \mathbf{\Sigma}\, \mathbf{C}^{\text{T}} + \mathbf{J}^{-1}(\overline{\alpha}\, B)|}{|\mathbf{J}^{-1}(\overline{\alpha}\, B)|} \right], \tag{19}$$

where α is our mixing parameter which may be varied between 0 and 1 to allocate the complementary sub-bands of B, and $\overline{\alpha} = 1 - \alpha$.

B. Successive Interference Cancellation

As in [1], we now look at the possibility of predicting and subtracting the radar return, decoding the communications signal to remove from the original return, and performing radar processing. This is known as the successive interference cancellation (SIC) bound. The composite radar/communications received signal minus our predicted radar return is given by:

$$\begin{aligned}\widetilde{z}(t) = &\sqrt{P_{\text{com}}}\, b\, s_{\text{com}}(t) + n(t) \\ &+ \sqrt{P_{\text{rad}}}\, a\, s_{\text{rad}}(t-\tau)\, e^{j\,\omega_{\text{D}}(t-\tau)} \\ &- \sqrt{P_{\text{rad}}}\, a\, s_{\text{rad}}(t-\tau_{\text{pre}})\, e^{j\,\omega_{\text{D,pre}}\left(t-\tau_{\text{pre}}\right)},\end{aligned} \tag{20}$$

where τ_{pre} and $\omega_{\text{D,pre}}$ are the predicted time delay and Doppler shift of our radar return respectively, and $s_{\text{com}}(t)$ is the communications waveform we wish to decode. Exploiting independence in the receiver noise $n(t)$, we can write the interference plus noise from the communications receiver's perspective as:

$$\begin{aligned}\sigma^2_{\text{int+n}} &= \text{E}\left[\left\| \widetilde{z}(t) - \sqrt{P_{\text{com}}}\, b\, s_{\text{com}}(t) \right\|^2 \right] \\ &= \sigma^2_{\text{noise}} + P_{\text{rad}}\, a^2\, \text{E}\left[\|\Delta s\|^2 \right],\end{aligned}$$

where Δs is the interference residual from the predicted subtraction. We can assume for simplicity that our predicted Doppler shift is approximately equivalent to the actual Doppler shift. For slowly accelerating targets in the middle of a track, this is an accurate assumption. Therefore

$$e^{j\,\omega_{\text{D,pre}}\left(t-\tau_{\text{pre}}\right)} \approx e^{j\,\omega_{\text{D}}\left(t-\tau+n_{\tau,\text{pre}}\right)} = e^{j\,\omega_{\text{D}}(t-\tau)}\, e^{j\,\omega_{\text{D}}\, n_{\tau,\text{pre}}} \tag{21}$$

since $\tau = \tau_{\text{pre}} + n_{\tau,\text{pre}}$ where $n_{\tau,\text{pre}}$ is the measurement process noise for the delay. However, the first term is the correct Doppler shift with the correct delay. Since we can pull this out of the residual, the expected residual is unaffected. Therefore:

$$\begin{aligned}&\text{E}\left[\|\Delta s\|^2 \right] = \\ &\text{E}\left[\left\| s_{\text{rad}}(t-\tau) - s_{\text{rad}}(t-\tau+n_{\tau,\text{pre}}) e^{j\,\omega_{\text{D}}\, n_{\tau,\text{pre}}} \right\|^2 \right].\end{aligned} \tag{22}$$

The L^2 norm produces 3 unique products, each of which we integrate over our CPI and take their expected value:

$$\begin{aligned}&\|s_{\text{rad}}(t-\tau)\|^2 \;,\; \|s_{\text{rad}}(t-\tau+n_{\tau,\text{pre}})\|^2 \;, \\ &-2\Re\left\{ s_{\text{rad}}(t-\tau) s^*_{\text{rad}}(t-\tau+n_{\tau,\text{pre}}) e^{-j\,\omega_{\text{D}}\, n_{\tau,\text{pre}}} \right\}\end{aligned} \tag{23}$$

The first two terms, integrated over our coherent processing interval, produce the signal RMS squared times our CPI

$$\|s_{\text{rad}}(t-\tau)\|^2 = s^2_{\text{rms}} T. \tag{24}$$

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We thus must solve for

$$A = \int_{-T/2}^{T/2} s_{\text{rad}}^*(t-\tau)\ \mathrm{E}\left[s_{\text{rad}}(t-\tau+n_{\tau,\text{pre}})e^{j\,\omega_\text{D}\,n_{\tau,\text{pre}}}\right] dt, \tag{25}$$

where we have pulled out the deterministic radar signal cross-term from the expectation and defined A to be the integral expectation of the last term in Equation (23). Note we have a random process governed by a Gaussian time delay and a Gaussian phase shift. To solve for this expectation, we can use Parseval's theorem:

$$A = \frac{1}{2\pi}\int_{-\infty}^{\infty} |S_{\text{rad}}(j\omega)|^2\ \mathrm{E}\left[e^{j\,\omega_\text{D}\,n_{\tau,\text{pre}}} e^{j\,\omega\,n_{\tau,\text{pre}}}\right] d\omega. \tag{26}$$

We note that the argument of the exponential is Gaussian. Therefore, the exponential term has a log-normal distribution [13]. The expectation is then the mean of a log-normal, which is given by [13]

$$\mathrm{E}\left[e^{j\,n_{\tau,\text{pre}}(\omega_\text{D}+\omega)}\right] = e^{-\frac{1}{2}(\omega_\text{D}+\omega)^2\sigma_{\tau,\text{pre}}^2}, \tag{27}$$

for $\sigma_{\tau,\text{pre}}^2$ the variance of the delay process noise. If we look carefully, we see this is a Gaussian function. Therefore, we see that the statistical mismatch in delay due to the model process noise of our delay results in a Gaussian spectral mask $G(j\omega)$ centered at our Doppler frequency. Therefore

$$\mathrm{E}\left[\|\Delta s\|^2\right] \approx 2s_{\text{rms}}^2\,T - \frac{1}{\pi}\int_{-\infty}^{\infty} |S_{\text{rad}}(j\omega)|^2\,G(j\omega)d\omega. \tag{28}$$

Note that we can make the simplifying assumption that our spectrum is flat, in which case we are simply integrating a non-normalized, truncated Gaussian. Finally,

$$\mathrm{E}\left[\|\Delta s\|^2\right] \approx 2s_{\text{rms}}^2\,T \left[1 - \frac{\mathrm{Q}\left(\omega_\text{D}\,\sigma_{\tau,\text{pre}} - B\,\pi\,\sigma_{\tau,\text{pre}}\right) - \mathrm{Q}\left(\omega_\text{D}\,\sigma_{\tau,\text{pre}} + B\,\pi\,\sigma_{\tau,\text{pre}}\right)}{B\sqrt{2\pi\,\sigma_{\tau,\text{pre}}^2}}\right]$$

where $\mathrm{Q}(x)$ is the complementary cumulative distribution function of the standard normal distribution. The communications rate is therefore defined for SIC as

$$R_{\text{com;SIC}} \le B\,\log_2\left[1 + \frac{b^2\,P_{\text{com}}}{k_\text{B}\,T_{\text{temp}}\,B + P_{\text{rad}}\,a^2\,\mathrm{E}\left[\|\Delta s\|^2\,;B\right]}\right] \tag{29}$$

where we have parameterized the residual power $\mathrm{E}\left[\|\Delta s\|^2\right]$ by the bandwidth. That is, the rate must be decreased due to the interference of the radar subtraction residual, which is a function of the target entropy. Since we can only predict the return, on average, as well as it is modeled, we have an average interference term. Assuming the communications system reduces its rate to handle the increase in noise and interference from the subtraction residual, we assume we can then decode the communications signal perfectly and remove it from the composite return so that the radar signal can then be processed as if it were the only system in band. Therefore

$$R_{\text{est;SIC}} \le \frac{1}{2T}\log_2\left[\frac{|\mathbf{C}\,\mathbf{\Sigma}\,\mathbf{C}^\mathrm{T} + \mathbf{J}^{-1}(B)|}{|\mathbf{J}^{-1}(B)|}\right]. \tag{30}$$

C. Water-filling

We can combine the concepts of both the ISB and SIC inner bounds to form another inner bound using water-filling (WF). The technique is described in detail in [1], and so we only present the final equation here for the sake of brevity. The basic idea is to allocate one sub-band of our full bandwidth B for communications only, and another for joint radar-communications use (using SIC). The sub-band allocation is parameterized by α as with the ISB, and for each allocation, an optimal communications power allocation to the communications only band and the mixture band is found through the general solution in [1]. The WF inner bound rates are therefore

$$R_{\text{com;com}} = \alpha B\,\log_2\left[1 + \frac{b^2\,\beta\,P_{\text{com}}}{k_\text{B}\,T_{\text{temp}}\,\alpha\,B}\right] \tag{31}$$

$$R_{\text{com;mix}} = \overline{\alpha}\,B\,\log_2\left[1 + \frac{b^2\,\overline{\beta}\,P_{\text{com}}}{k_\text{B}\,T_{\text{temp}}\,\overline{\alpha}\,B + P_{\text{rad}}\,a^2\,\mathrm{E}\left[\|\Delta s\|^2\,;\overline{\alpha}\,B\right]}\right] \tag{32}$$

$$R_{\text{com;WF}} \le R_{\text{com;com}} + R_{\text{com;mix}} \tag{33}$$

$$R_{\text{est;WF}} \le \frac{1}{2T}\log_2\left[\frac{|\mathbf{C}\,\mathbf{\Sigma}\,\mathbf{C}^\mathrm{T} + \mathbf{J}^{-1}(\overline{\alpha}\,B)|}{|\mathbf{J}^{-1}(\overline{\alpha}\,B)|}\right] \tag{34}$$

where $\overline{\beta} = 1-\beta$. The optimal power distribution β is given by [10]

$$\beta = \begin{cases} \alpha + \frac{1}{P_{\text{com}}}\left(\frac{\overline{\alpha}}{\mu_{\text{com}}} + \frac{\alpha}{\mu_{\text{mix}}}\right) & P_{\text{com}} \ge \frac{\alpha}{\overline{\alpha}\,\mu_{\text{mix}}} - \frac{1}{\mu_{\text{com}}} \\ 1 & P_{\text{com}} < \frac{\alpha}{\overline{\alpha}\,\mu_{\text{mix}}} - \frac{1}{\mu_{\text{com}}} \end{cases} \tag{35}$$

for

$$\mu_{\text{com}} = \frac{b^2}{k_\text{B}\,T_{\text{temp}}\,\alpha\,B}, \text{ and} \tag{36}$$

$$\mu_{\text{mix}} = \frac{b^2}{k_\text{B}\,T_{\text{temp}}\,\overline{\alpha}\,B + P_{\text{rad}}\,a^2\,\mathrm{E}\left[\|\Delta s\|^2\,;B\right]}. \tag{37}$$

IV. Examples

In Figure 2, we see an example of all the bounds we have extended in this paper. We used parameters similar to those used in [1] for comparison, as shown in Table I. The communications SIC vertex is shown linearly interpolated to

TABLE I
PARAMETERS FOR EXAMPLE PERFORMANCE BOUND

Parameter	Value	Parameter	Value
Bandwidth	5 MHz	$\omega_c/2\pi$	3 GHz
Abs. Temperature	1000 K	Comms Range	10 km
Comms Power	100 W	Comms Antenna Gain	10 dBi
Prev. Target Range	40 km	Radar Antenna Gain	30 dBi
Radar Power	100 kW	Target Cross Section	100 m^2
CPI	250 ms	Process Noise Intensity	250000
RMS BW	$B/\sqrt{12}$	RMS Envelope	$T/\sqrt{12}$
EM Propagation	$3{\cdot}10^8$ m/s	Prev. Bearing	22.5°
Prev. x Speed	125 m/s	Prev. y Speed	125 m/s

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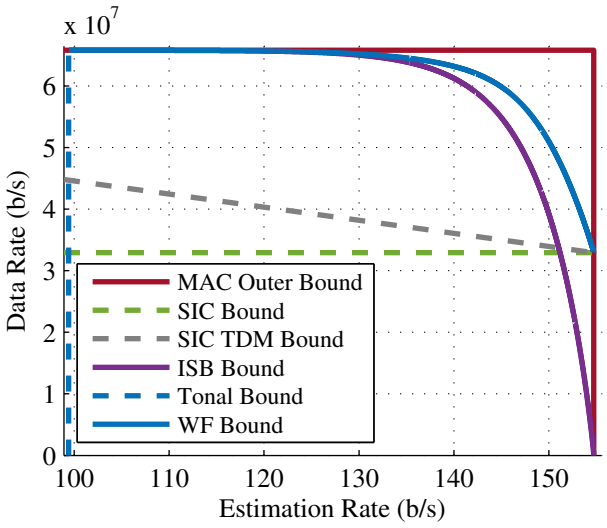


Fig. 2. Data rate and estimation rate bounds. The multiple access channel (MAC) outer bound is shown in the solid red lines, indicating the independent rates for each system. The isolated sub-band (ISB) inner bound is shown by the solid purple line. The range-rate only (tonal radar) estimation rate is shown by the dashed blue line. The successive interference cancellation (SIC) bound is shown by the dashed green line, and the joint performance achieved by time division multiplexing (TDM) of the radar and the communications operating with SIC is shown by the dashed gray line. Finally, the water-filling (WF) inner bound is shown in solid blue.

the radar-free communications bound. This is achievable because we can simply utilize time-division multiplexing (TDM) between operating at the SIC vertex with joint radar and communications, and simply operating the communications by itself [3]. It appears both the ISB and WF inner bounds approach an asymptote for the radar estimation rate. This is the tonal bound, defined as the point at which the bandwidth vanishes, and only a tone is used for the radar. As a result, only the range-rate measurement contributes to the joint mutual information. As we can see, the WF bound outperforms all of the other inner bounds for this parameter set, though it may not in general. The estimation rate also appears much lower than in [1], but this is due to the long CPI with a similar source entropy. Note we have started the estimation rate axis at the tonal bound to emphasize detail on the same order of magnitude as in [1].

V. Conclusion

In this paper we extended the previous results for producing joint radar-communications performance bounds to accommodate FMCW radars. This extension necessitated a more thorough development of a Markov motion model, as well as an extended measurement model to include Doppler estimation. This in turn required a reformulation of the SIC inner bound statistical interference term due to Doppler phase mismatch. Even with these generalizations, there are a large range of significant extensions to these bounds to make them applicable to more scenarios. While information has not historically been considered a figure of merit for radar, with cognitive radar architectures taking center stage [16], information-centric measures are needed to intelligently control radar emissions [17] and maximize spectral cooperation with communications users.

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